

Review for Midterm 1 Math 2270

Some examples/solutions

- ① Find a system of equations with solutions $\begin{pmatrix} 2 \\ 1 \\ -1 \end{pmatrix} + t \begin{pmatrix} 1 \\ 1 \\ 1 \end{pmatrix}$

Answer We need $2 = 3 - 1$ equations.

The null space is $\begin{pmatrix} 1 \\ 1 \\ 1 \end{pmatrix}$ so the matrix has 2 pivots

and should look like $\begin{pmatrix} 1 & 0 & * \\ 0 & 1 & * \end{pmatrix}$ so $\begin{pmatrix} 1 & 0 & -1 \\ 0 & 1 & -1 \end{pmatrix}$

The augmented matrix looks like $\left[\begin{array}{ccc|c} 1 & 0 & -1 & b_1 \\ 0 & 1 & -1 & b_2 \end{array} \right]$

plugging in $\begin{pmatrix} 2 \\ 1 \\ -1 \end{pmatrix}$ (the special solution) we get $b_1 = 3$
 $b_2 = 2$

$$\text{so } \begin{cases} x - z = 3 \\ y - z = 2 \end{cases}$$

- ② Find a system of eqns with sols $\begin{pmatrix} 1 \\ 1 \\ 1 \end{pmatrix} + s \begin{pmatrix} 1 \\ 1 \\ 0 \end{pmatrix} + t \begin{pmatrix} 0 \\ 1 \\ 1 \end{pmatrix}$

We need 1 eqn so the matrix looks like $\boxed{1 \ * \ *}$

since $\begin{pmatrix} 1 \\ 1 \\ 0 \end{pmatrix}$ is in the null space we get $\boxed{1 \ -1 \ *}$

since $\begin{pmatrix} 0 \\ 1 \\ 1 \end{pmatrix}$ is in the null space we get $\boxed{1 \ -1 \ 1}$

$$\text{so get } x - y + z = b \Rightarrow 1 - 1 + 1 = b \Rightarrow b = 1$$

$$\underline{\underline{x - y + z = 1}}$$

(3) Find system of eqns with sol $\begin{pmatrix} 1 \\ 0 \\ 0 \end{pmatrix} + t \begin{pmatrix} 3 \\ 1 \\ 0 \\ 2 \end{pmatrix}$

Sol $n = P + F$ so $P = 4 - 1 = 3$ need 3 eqns $\begin{pmatrix} 1 \\ 0 \\ 0 \end{pmatrix}$

try $\begin{bmatrix} 1 & 0 & 0 & x \\ 0 & 1 & 0 & y \\ 0 & 0 & 1 & z \end{bmatrix}$ so $\begin{bmatrix} 1 & 0 & 0 & -3/2 \\ 0 & 1 & 0 & -1/2 \\ 0 & 0 & 1 & 0 \end{bmatrix}$

need equations $\begin{cases} x - 3/2 u = b_1 \\ y - 1/2 u = b_2 \\ z = b_3 \end{cases}$ plug in $\begin{pmatrix} 1 \\ 0 \\ 1 \\ 0 \end{pmatrix}$ to get

$$\begin{cases} x - \frac{3}{2} u = 1 \\ y - \frac{1}{2} u = 0 \\ z = 1 \end{cases}$$

(4) Sols of $\begin{bmatrix} 1 & 2 & 0 & 3 & 5 \\ 1 & 2 & 1 & 5 & 7 \end{bmatrix} \begin{pmatrix} x \\ y \\ z \\ u \\ v \end{pmatrix} = \begin{pmatrix} 5 \\ 8 \end{pmatrix}$

$\begin{bmatrix} 1 & 2 & 0 & 3 & 5 & | & 5 \\ 1 & 2 & 1 & 5 & 7 & | & 8 \end{bmatrix} \rightarrow \begin{bmatrix} 1 & 2 & 0 & 3 & 5 & | & 5 \\ 0 & 0 & 1 & 2 & 2 & | & 3 \end{bmatrix}$ part sol $\begin{pmatrix} 5 \\ 0 \\ 3 \\ 0 \\ 0 \end{pmatrix}$ special sols $\begin{matrix} -2 & -3 & -5 \\ 0 & 0 & 0 \\ 0 & -2 & -2 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{matrix}$

so $\begin{pmatrix} 5 \\ 0 \\ 3 \\ 0 \\ 0 \end{pmatrix} + s \begin{pmatrix} -2 \\ 1 \\ 0 \\ 0 \\ 0 \end{pmatrix} + t \begin{pmatrix} -3 \\ 0 \\ -2 \\ 1 \\ 0 \end{pmatrix} + u \begin{pmatrix} -5 \\ 0 \\ -2 \\ 0 \\ 1 \end{pmatrix}$

(4) $\begin{bmatrix} 1 & 2 & 0 \\ 2 & 1 & 2 \\ 0 & c & 1 \end{bmatrix} \rightarrow \begin{bmatrix} 1 & 2 & 0 \\ 0 & -3 & 2 \\ 0 & c & 1 \end{bmatrix} \rightarrow \begin{bmatrix} 1 & 2 & 0 \\ 0 & -3 & 2c \\ 0 & c & 1 \end{bmatrix}$

3 pivots unless
 $-3 - 2c = 0$
 $-3 = 2c$
 $c = -3/2$

If $c \neq -3/2$ $\text{rk} = 3$ $N(A) = 0$

If $c = -3/2$ $\text{rk} = 2$ $\begin{bmatrix} 1 & 2 & 0 \\ 0 & -3/2 & 1 \end{bmatrix} \rightarrow \begin{bmatrix} 1 & 0 & 4/3 \\ 0 & 1 & -2/3 \end{bmatrix}$ $N(A) = \left\langle \begin{pmatrix} -4/3 \\ 2/3 \\ 1 \end{pmatrix} \right\rangle$

Ex For which b_1, b_2, b_3 can we solve:

$$\left[\begin{array}{ccc|c} 1 & 4 & 2 & b_1 \\ 2 & 8 & 4 & b_2 \\ -1 & -4 & -2 & b_3 \end{array} \right] \rightarrow \left[\begin{array}{ccc|c} 1 & 4 & 2 & b_1 \\ 0 & 0 & 0 & b_2 - 2b_1 \\ 0 & 0 & 0 & b_3 + b_1 \end{array} \right] \quad \text{need } \begin{array}{l} b_2 - 2b_1 = 0 \\ b_3 + b_1 = 0 \end{array} \Rightarrow \begin{array}{l} b_2 = 2b_1 \\ b_3 = -b_1 \end{array}$$

$$\left[\begin{array}{ccc|c} 1 & 4 & 2 & b_1 \\ 2 & 8 & 4 & b_2 \\ -1 & -4 & -2 & b_3 \end{array} \right] \rightarrow \left[\begin{array}{ccc|c} 1 & 4 & 2 & b_1 \\ 0 & 1 & 0 & b_2 - 2b_1 \\ 0 & 0 & 0 & b_3 + b_1 \end{array} \right] \quad b_3 + b_1 = 0 \text{ unique solution}$$

$$\left[\begin{array}{ccc|c} 1 & 0 & 2 & b_1 \\ 0 & 1 & 0 & b_2 \\ 0 & 0 & 1 & b_3 \end{array} \right] \quad \text{any } b_i \text{ are ok}$$

$$\rightarrow \left[\begin{array}{ccc|c} 1 & 0 & 0 & b_1 - 2b_3 \\ 0 & 1 & 0 & b_2 \\ 0 & 0 & 1 & b_3 \end{array} \right]$$

unique sol is $\begin{array}{l} x = b_1 - 2b_3 \\ y = b_2 \\ z = b_3 \end{array}$

Ex $1 \cdot \begin{pmatrix} 1 \\ 0 \\ 1 \end{pmatrix} + 1 \cdot \begin{pmatrix} 0 \\ 1 \\ 1 \end{pmatrix} - 1 \cdot \begin{pmatrix} 1 \\ 1 \\ 2 \end{pmatrix} = \begin{pmatrix} 0 \\ 0 \\ 0 \end{pmatrix}$ so $\begin{pmatrix} 1 \\ 0 \\ 1 \end{pmatrix}, \begin{pmatrix} 0 \\ 1 \\ 1 \end{pmatrix}, \begin{pmatrix} 1 \\ 1 \\ 2 \end{pmatrix}$ are LI

$$\left[\begin{array}{ccc} 1 & 0 & 0 \\ 0 & 1 & 1 \\ 1 & 1 & 2 \end{array} \right] \rightarrow \left[\begin{array}{ccc} 1 & 0 & 0 \\ 0 & 1 & 1 \\ 0 & 0 & 1 \end{array} \right] \text{ has max rk so } \begin{pmatrix} 1 \\ 0 \\ 1 \end{pmatrix}, \begin{pmatrix} 0 \\ 1 \\ 1 \end{pmatrix}, \begin{pmatrix} 0 \\ 1 \\ 2 \end{pmatrix} \text{ are LI}$$

Ex $a + bx + cx^2 = 0 \Rightarrow \begin{array}{l} b + 2cx = 0 \text{ (first der)} \\ 2c = 0 \text{ (2nd der)} \end{array}$

So $c = 0$ but then $b = 0$ & $a = 0$

So $1, x, x^2$ are LI

$1 \cdot \cos^2 x + 1 \cdot \sin^2 x - 1 = 0$ so $\cos^2 x, \sin^2 x, 1$ are LI

$a \cos x + b \sin x = 0 \Rightarrow \begin{array}{l} a \cos 0 + b \sin 0 = 0 \Rightarrow a = 0 \\ a \cos \pi/2 + b \sin \pi/2 = 0 \Rightarrow b = 0 \end{array} \Rightarrow \text{LI}$

Ex Let $S = \{A \in M \mid A = A^T\}$ be the set of symmetric matrices then S is a subspace of M .

proof (1) $0^T = 0$ so $0 \in S$ ✓

(2) $A, B \in S \Rightarrow A^T = A$ & $B^T = B \Rightarrow (A+B)^T = A^T + B^T = A+B$

so $A+B \in S$ ✓

(3) $A \in S \Rightarrow A^T = A$ $\Rightarrow (\lambda A)^T = \lambda^T A^T = \lambda A^T$
 $\lambda \in \mathbb{R} \quad \lambda^T = \lambda \quad \Rightarrow \lambda A \in S$ ✓

$$\text{Ex } N \begin{pmatrix} 1 & 2 & 0 & 3 & 5 \\ & 1 & 2 & 1 & 5 & 7 \end{pmatrix} = N \begin{pmatrix} 1 & 2 & 0 & 3 & 5 \\ & 0 & 0 & 1 & 2 & 2 \end{pmatrix} = \begin{pmatrix} -2 & -3 & -5 \\ 1 & 0 & 0 \\ 0 & -2 & -2 \\ 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{pmatrix}$$

$$N \begin{pmatrix} 1 & 2 & 3 \\ 1 & 0 & 4 \\ 0 & 2 & -1 \end{pmatrix} = N \begin{pmatrix} 1 & 2 & 3 \\ 0 & -2 & 1 \\ 0 & 0 & 0 \end{pmatrix} = N \begin{pmatrix} 1 & 2 & 3 \\ 0 & 1 & -1/2 \\ 0 & 0 & 0 \end{pmatrix}$$

$$= N \begin{pmatrix} 1 & 0 & 2 \\ 0 & 1 & -1/2 \\ 0 & 0 & 0 \end{pmatrix} = \begin{pmatrix} -2 \\ 1/2 \\ 1 \end{pmatrix}$$

$$N \begin{pmatrix} 1 & 0 & 3 \\ 0 & 1 & 2 \\ 2 & 2 & 1 \\ 1 & 1 & 5 \end{pmatrix} = N \begin{pmatrix} 1 & 0 & 3 \\ 0 & 1 & 2 \\ 0 & 2 & 5 \\ 0 & 1 & 2 \end{pmatrix} = N \begin{pmatrix} 1 & 0 & 3 \\ 0 & 1 & 2 \\ 0 & 0 & 1 \\ 0 & 0 & 0 \end{pmatrix} = \begin{pmatrix} 0 \\ 0 \\ 0 \end{pmatrix}$$